

5 (**short**) A small factory consumes 24 kW of power with a lagging power factor of 0.7 at the factory. When it is connected to a 50 Hz supply, the rms voltage across the factory terminals is 240 V. The supply line has an impedance $Z_s = (0.02 + j0.04) \Omega$.

(a) Draw a circuit diagram and calculate the real power lost in the line, and the voltage supplied at the input. [6]

(b) Assuming that the rms voltage across the factory terminals continues to be 240 V, calculate the value of the capacitor connected across the factory terminals that is needed to correct its power factor to unity. [4]